



Full length article

Efficient object tracking on edge devices with MobileTrack[☆]Jiang Zhai^a, Zinan Cheng^b, Wenkang Zhang^a, Dejun Zhu^{c,*}, Wankou Yang^{a,*}^a School of Automation, Southeast University, Nanjing 210096, China^b Jiangsu Automation Research Institute, Lianyungang 222061, China^c State Key Laboratory of Hydrosience and Engineering, Department of Hydraulic Engineering, Tsinghua University, Beijing 100084, China

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ABSTRACT

Object tracking has made significant progress in recent years. However, the state-of-the-art trackers are becoming increasingly heavy and expensive, making their deployment challenging in resource-constrained applications. In this study, we introduce MobileTrack, a visual object tracker that strikes a perfect balance between tracking accuracy and inference speed. Utilizing a novel coordinated perception-aware fusion module and a lightweight prediction head, our proposed methodology outperforms most Siamese trackers on various academic benchmarks in terms of both accuracy and efficiency. When deployed on resource-constrained embedded devices such as NVIDIA Jetson TX2, MobileTrack ensures real-time performance at a speed exceeding 33FPS, while LightTrack only operates at 18FPS. Therefore, MobileTrack holds significant potential to unlock a wide range of practical applications across various industries. MobileTrack is released at [here](#).

1. Introduction

Object tracking is a fundamental and important research branch of computer vision. In recent years, with the continuous development of deep learning technology, object tracking has made significant progress and the tracking accuracy has also been continuously improved. However, this has also caused complex and expensive computations, which has brought many problems to the deployment of object tracking algorithms. Especially with the rise of Transformer [1] architecture, object tracking has benefited from Transformer's long-term dependence, and the accuracy and robustness have been greatly improved. But the complex self-attention mechanism of Transformer makes related algorithms far from real-time. For example, STARK-S50 [2] has an inference speed of 40 FPS on Tesla V100 GPU, while the model inference speed on NVIDIA Jetson TX2 is only 5.56 FPS. The high computation costs associated with tracking models pose a significant challenge for their practical deployment in real-world applications, including drones, industrial robots, and edge embedded devices for unmanned driving. As such, there is a pressing need to develop robust and efficient tracking algorithms that can overcome these limitations. The pursuit of novel methods that strike a balance between accuracy and speed remains a crucial research direction in the field of visual tracking, as it holds the potential to unlock a range of practical applications across various industries.

Currently, cutting-edge methodologies in visual object tracking exhibit promising performance on publicly accessible datasets, including

the Visual Object Tracking (VOT) challenges [5] and the Online Tracking Benchmark (OTB) [6]. However, these achievements come at a significant computational cost, necessitating substantial computational resources for implementation. In prior research, three primary approaches have been employed to develop lightweight models. The first approach involves utilizing compression techniques, such as pruning and quantization, to reduce the computational burden of the models [7]. Current off-the-shelf compression techniques, such as pruning and quantization, aim to diminish model complexity. However, they inherently introduce non-negligible performance degradation attributable to information loss [8,9]. The second approach involves manually designing efficient and compact models, such as FEAR [10], with the goal of minimizing computational overhead. Lastly, the third approach involves leveraging Neural Architecture Search (NAS) techniques to automatically design lightweight models, as exemplified by LightTrack [4]. However, the count of FLOP does not consistently depict the real inference time.

We adhered to established best practices in formulating an efficient and adaptable neural network architecture. The present study presents a lightweight visual object tracker, named MobileTrack, for edge devices. Drawing upon existing research in the field of object tracking, we manually devise a lightweight tracking framework that aims to strike a delicate balance between tracking accuracy and inference speed. The proposed methodology contributes to the growing body of literature on object tracking for edge computing, offering a

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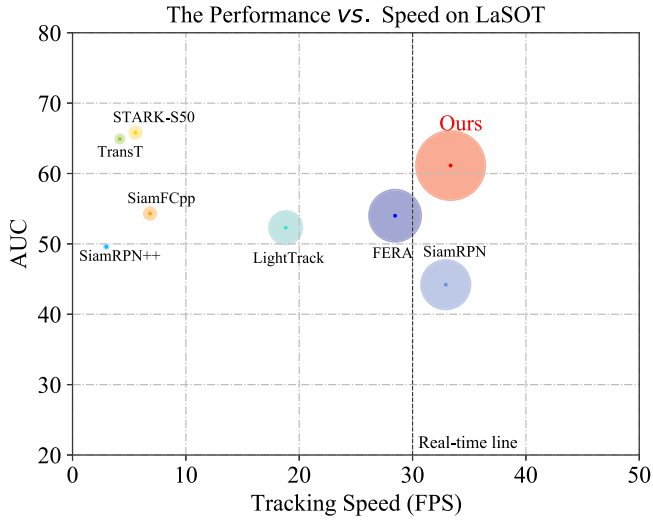


Fig. 1. Comparison of tracker performance on LaSOT [3], showing AUC scores (Success in %) and corresponding tracking speeds in fps on a standard NVIDIA Jetson TX2. Our novel MobileTrack tracker is shown to outperform all other real-time trackers, exhibiting an impressive 8.8% higher AUC score than the mobile LightTrack [4]. Our proposed algorithm achieves an optimal balance between accuracy and speed, making it a promising solution for practical applications that demand efficient and precise visual tracking.

practical solution that can be deployed on resource-constrained devices. Taking into account the existing body of research, we have developed a novel lightweight visual object tracking model, MobileTrack, which is specifically designed for deployment on edge devices. By striking a balance between tracking accuracy and inference speed, our proposed algorithm is well-suited for real-time visual tracking applications in resource-constrained environments. Through careful design and optimization, our approach offers significant gains over previous methods in terms of computational efficiency, while still maintaining high levels of tracking performance (see Fig. 1). The results of our study demonstrate the potential of MobileTrack to address the challenges of visual tracking in edge computing scenarios, advancing the capabilities of these devices for a range of practical applications.

The main contributions of this paper are as follows:

- We present a new fusion module that enhances perception coordination by incorporating position information into channel attention mechanisms. Our method encodes both channel relations and spatial details, facilitating thorough fusion of template and search frames for effective relationship modeling. Compared to other fusion methods, we achieve seamless integration of templates and search with reduced computational costs.
- We developed a lightweight visual object tracker named MobileTrack, striking an optimal balance between tracking accuracy and speed. In LaSOT testing, MobileTrack outperformed other real-time trackers with an AUC score of **61.1**, 8.8% higher than LightTrack. Additionally, MobileTrack achieved fast inference speeds exceeding **33 FPS** on the NVIDIA Jetson TX2 platform. These results highlight MobileTrack's potential for reliable and efficient visual tracking in diverse real-world scenarios.

2. Related work

2.1. Visual object tracking

Early single-object tracking algorithms borrowed the correlation filtering algorithm using Fast Fourier Transform (FFT), mainly using

filters to combine position responses and tracking results, and repeatedly training models to achieve accurate positioning [11–13]. In recent years, Siamese networks [14–21] and discriminatory correlation filters [22,23] have become the mainstream of the Visual Object Tracking.

The classic tracking algorithm, SiamFC [14], is widely recognized as a pivotal contribution to the field of target tracking, having laid the foundation for the use of Siamese networks in this area. Building on this foundation, subsequent research has yielded several significant advancements. SiamRPN [15], for instance, combines the Siamese network with RPN [24], enabling multi-scale tracking capabilities. SiamRPN++ [16] takes this further by utilizing a deeper network to fully exploit the advantages of deep features. Meanwhile, SiamMask [17] extends the capabilities of Siamese trackers by not only obtaining target tracking boxes but also determining target boundaries and completing instance segmentation.

Inspired by FCOS [25] in object detection, both SiamBAN [18] and SiamCAR [19] adopt the anchor-free strategy. Unlike preset anchor boxes, this approach provides the tracker with greater degrees of freedom. Another notable advancement is observed in SiamFC++ [20], which incorporates the anchor-free strategy and integrates position and quality scores, resulting in a significant improvement in tracking performance. To overcome the requirement of predefining the target feature region size, SiamGAT [21] utilizes the graph attention mechanism to propagate target information from template features to search features. In addition to Siamese-based trackers, other notable approaches have leveraged discriminative correlation filters with online updating to achieve discriminative target trackers, such as ATOM [22] and DiMP [23]. Overall, these contributions demonstrate the ongoing refinement and evolution of visual tracking algorithms towards more reliable and accurate performance.

To enhance the modeling of the relationship between the template frame and the search frame in the context of visual object tracking, the application of the Transformer has been introduced. For instance, TransT [26] leverages multiple self-attention layers and cross-attention layers to achieve feature fusion. Similarly, STARK [2] connects template features and region features, and feeds them into multiple self-attention heads, resulting in improved performance. OSTRack [27] adopts a one-stream one-stage tracking framework that enables simultaneous feature extraction and relationship modeling. Transformer-based trackers have exhibited notable advancements in tracking performance as compared to Siamese Trackers, owing to their ability to establish long-term dependencies. Consequently, Transformer-based algorithms have emerged as the most prominent and optimal mainstream approach in the field of single-object tracking.

The surge in demand for real-time visual tracking, particularly in fields like autonomous vehicles and human-computer interaction, emphasizes the significance of efficient deep tracking architectures. Trackers like KCF [12], utilizing handcrafted features, operate in real-time on a CPU. Despite their speed, their dependence on handcrafted features substantially compromises accuracy and robustness, leading to suboptimal performance when compared to more recent and intricate methods. LightTrack [4] utilizes NAS to identify a Siamese tracking architecture that is both lightweight and efficient. In a similar vein, FEAR [10] adopts a manual design strategy, crafting a tracker with low computational demands by incorporating a succinct feature extraction network, employing dual-template representation, and implementing blocks for pixel-wise fusion.

2.2. Efficient neural networks

With the increasing demand for inference optimization of neural networks on embedded and mobile devices, efficient and lightweight neural networks have received widespread attention. MobileNet [28] is one of the most popular lightweight network structures. It uses depthwise separable convolution to reduce computation and 1×1

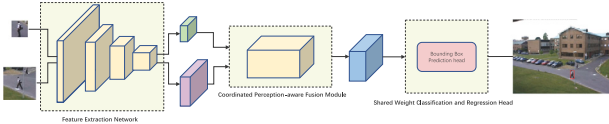


Fig. 2. The MobileTrack network architecture is comprised of three essential components: a lightweight feature extraction network, a coordinated perception-aware fusion module, and a shared weight classification and regression head. The backbone of the model is responsible for extracting feature representations from both the template and search images, while the feature fusion module plays a crucial role in effectively combining these features. Finally, the bounding box prediction head is responsible for making the final predictions regarding the location of the bounding box and its presence in the video frames.

convolution to transform the number of channels to reduce model complexity. MobileNetV2 [29] is a further improvement of MobileNet. In order to further reduce computation and improve accuracy, MobileNetV2 uses new technologies such as Inverted Residuals and Linear Bottlenecks. Through these technologies, MobileNetV2 reduces more computation and model parameters while maintaining or even surpassing the accuracy of MobileNetV1. Compared with MobileNetV1, MobileNetV2 is more optimized in structure, therefore it has greater advantages in training speed and model size. ShuffleNet [30] is another common lightweight network structure. It uses group convolution and channel shuffling mechanism to reduce model parameters and computation. Group convolution divides input and output channels into groups that are independent of each other to reduce computation. Channel shuffling mechanism increases model diversity and reduces computation by grouping and cross-reusing channels. SqueezeNet [31] is a lightweight network designed to compress network volume. It uses Fire Module to reduce network parameters. The Fire Module consists of a Squeeze layer and an Expand layer. The Squeeze layer reduces input channels by using 1×1 convolution, while the Expand layer expands channels back to their original dimensions using 1×1 and 3×3 convolution. GhostNet [32] reduces the number of parameters and computation in the network by using ghost modules to achieve lightweight goals through simple operations that obtain more feature maps.

The emergence of Vision Transformer has established the Transformer as a mainstream approach in the field of computer vision. However, like natural language understanding, Transformers in computer vision face challenges stemming from their large parameter spaces and challenging deployment procedures. Consequently, there is a growing need for lightweight Transformers that can efficiently tackle downstream tasks. To address this need, MobileViT [33] combines the capabilities of Convolutional Neural Networks (CNN) and Transformer to produce lightweight networks that fuse both approaches. By integrating the Transformer as a convolution component into MobileNetV2, MobileViT represents a promising advancement towards achieving efficient and effective solutions in the field of computer vision.

The development of these lightweight networks has brought more opportunities and challenges to research and applications in computer vision and other fields. In the future, further exploration by researchers is essential in this domain to offer more efficient and accurate solutions tailored to diverse edge devices and application scenarios.

3. Proposed approach

3.1. Overall pipeline

The MobileTrack tracker represents a fast and efficient single object tracking framework that comprises a lightweight feature extraction network, a coordinated perception-aware fusion module, and a shared weight classification and regression head. To optimize real-time performance, we avoid adopting a dynamic template strategy akin to

STARK-ST [2], instead relying on the initial frame's provided template image throughout the tracking process without updating it. To streamline the tracking process, MobileTrack extracts the template feature z_f by cropping the specified target area in the initial frame. Subsequently, it does not update this template feature in subsequent video frames. Next, the tracker extracts the search feature x_f from the search frame and utilizes the feature fusion network to perform cross-correlation operations between z_f and x_f , thereby establishing a connection between them. Finally, the fused features are fed into the head subnetwork to classify and regress the target's position, resulting in highly accurate tracking. Fig. 2 illustrates the entire architecture of the model, and each module will be analyzed separately.

3.2. Lightweight feature extraction network

In the realm of visual object tracking, there is a pressing need for lightweight and swift models. Specifically, such models necessitate feature extraction networks that are similarly lightweight. To achieve accurate target localization, it is crucial that the extracted features possess adequate resolution. In previous tracking approaches, exemplified by SiamRPN++, stride of the backbone is typically reduced to enhance output resolution. This is achieved by adjusting the stride of the last two stages of ResNet-50 to 1 and utilizing dilated convolutions to enhance the receptive fields. However, such methods often compromise real-time performance. Recent experimental results from FEAR have shown that maintaining the original spatial resolution can significantly decrease computational costs associated with both the backbone and prediction head, while still achieving commendable outcomes.

We leverage the initial four stages of MobileViT-S, pre-trained on ImageNet, as our backbone. To ensure a constant number of channels in the output feature map, an AdjustLayer is employed. This layer incorporates both Convolutional and Batch Normalization layers for depth mapping.

Similar to a Siamese tracking framework, the proposed MobileTrack consists of two branches: the template branch and the search branch. The former takes cropped image z of size $W_z \times H_z$ from the initial frame as a reference, while the latter takes the cropped image x of size $W_x \times H_x$ from the current frame for tracking. Notably, both inputs are processed by the same backbone network, which yields the template feature map $f_z \in \mathbb{R}^{\frac{H_z}{s} \times \frac{W_z}{s} \times C}$ and the search feature map $f_x \in \mathbb{R}^{\frac{H_x}{s} \times \frac{W_x}{s} \times C}$, where s denotes the spatial reduction ratios and C represents the number of channels in the feature maps. Specifically, for our research, we consider the values $s = 16$ and $C = 128$.

3.3. Coordinated perception-aware fusion module

The correlation module accomplishes relationship modeling between the template and search images, generating a unified representation. SiamFC employs simple cross-correlation operations while SiamRPN++ utilizes depth-wise convolution for relationship modeling. Recently, pixel-wise cross-correlation operations have been used by RANet [34] and Alpha-Refine [35] to model the relationship between templates and searches. Pixel correlation refers to measuring the similarity between two images at each pixel position. To fully integrate template and search frames for relationship modeling, we designed a coordinated perception-aware fusion module. Pixel-wise correlation is also employed to achieve pixel-level cross-correlation operations between f_z and f_x . In order to fully exploit the information contained in the feature map f_{corr} obtained through cross-correlation, the position information is embedded into channel attention. This allows for capturing of inter-channel relationships as well as long-range dependencies that possess precise position information. The Squeeze-and-Excitation (SE) [36] generally neglects the positional information, which is important for generating spatially selective attention maps. We overcome this limitation by capturing long-range dependencies along one

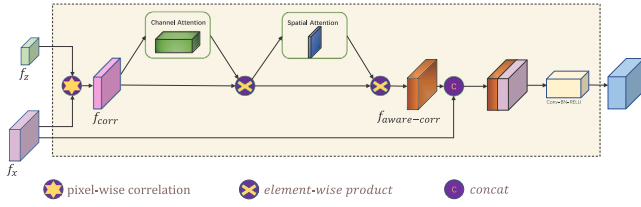


Fig. 3. The structure of the coordinated perception-aware fusion module.

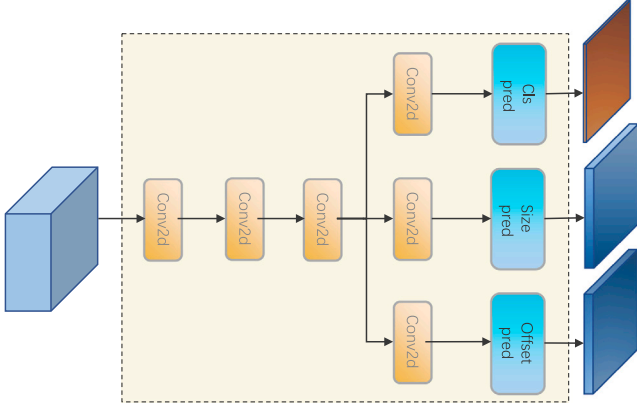


Fig. 4. The structure of the shared weight classification and regression head.

spatial direction while simultaneously retaining precise position information along another spatial direction. Thereafter, the generated feature maps are encoded independently into a pair of direction-sensitive and position-sensitive feature maps. The coordinated perception-aware fusion module is capable of encoding channel relationships and long-range dependencies with accurate position information. The specific structure of the module is illustrated in Fig. 3.

The template feature f_z and search feature f_x extracted from the feature extraction network are forwarded to the coordinated perception-aware fusion module for relationship modeling. Fast matrix multiplication is employed to establish pixel-level cross-correlation operation to determine the similarity between f_z and f_x , resulting in the feature map f_{corr} . Following this, channel attention and spatial attention operations are applied to f_{corr} , producing the refined feature map $f_{aware-corr}$. This enhanced map captures inter-channel relationships while maintaining precise location information. The feature map $f_{aware-corr}$ which has channel and spatial information, is concatenated with the search image feature f_x , and the resulting feature is aggregated through a 1×1 Conv-BN-RELU block. Utilizing the coordinated perception-aware fusion module results in learned features that have stronger discriminative ability and can effectively encode the position and appearance of objects in the tracker.

3.4. Shared weight classification and regression head

The feature maps derived from the Coordinated Perception-aware Fusion Module undergo further feature extraction through a fully convolutional network (FCN), which comprises L stacked Conv-BN-ReLU layers. This branch is responsible for obtaining both the classification and regression features needed for prediction, with a focus on minimizing parameter count. The prediction head is then split into three branches, each composed of a few Conv-BN-ReLU layers, as shown in Fig. 4.

The bounding box prediction head produces the final outputs, which consist of the target classification score map represented by $P \in [0, 1]^{\frac{H_x}{s} \times \frac{W_x}{s}}$, the local offset $O \in [0, 1]^{2 \times \frac{H_x}{s} \times \frac{W_x}{s}}$ utilized to compensate

for discretization errors arising from reduced resolution, and the normalized bounding box size $S \in [0, 1]^{2 \times \frac{H_x}{s} \times \frac{W_x}{s}}$. The position is identified by means of $(x_d, y_d) = \text{argmax}_{(x,y)} P$ and final target bounding box is obtained as:

$$\begin{cases} x = x_d + O(0, x_d, y_d), \\ y = y_d + O(1, x_d, y_d), \\ w = S(0, x_d, y_d), \\ h = S(1, x_d, y_d), \end{cases} \quad (1)$$

3.5. Loss function

In this study, we incorporate the weighted focal loss [37] for classification. Specifically, we utilize a Gaussian kernel to generate the ground truth heatmap, denoted as $\hat{P}$, based on a ground truth target center $\hat{p}$ and its corresponding low-resolution equivalent $\tilde{p} = [\tilde{p}_x, \tilde{p}_y]$. The Gaussian kernel expression is defined as $\hat{P} = \exp\left(-\frac{(x-\tilde{p}_x)^2 + (y-\tilde{p}_y)^2}{2\sigma^2}\right)$. It is noteworthy that the parameter σ signifies an object size-adaptive standard deviation. The Gaussian weighted focal loss is then formulated as:

$$L_{cls} = - \sum_{xy} \begin{cases} (1-P)^\alpha \log(P), & \text{if } \hat{P} = 1 \\ (1-\hat{P})^\beta P^\alpha \log(1-P), & \text{otherwise} \end{cases} \quad (2)$$

here, α and β denote hyper-parameters, and we set $\alpha = 2$ and $\beta = 4$, in accordance with previous work [27,37].

we adopt the ℓ_1 Loss and the generalized IoU loss [38] for bounding box regression with the predicted bounding box. The overall loss function is then formulated as a weighted combination of these three losses.

$$L = L_{cls} + \lambda_{iou} L_{iou} + \lambda_{L1} L_1 \quad (3)$$

In our experiments, we set the regularization parameters $\lambda_{iou} = 2$ and $\lambda_{L1} = 5$ in accordance with previous work [2].

4. Experiments

4.1. Implementation details

Training phase. In this study, we implement our models on the py-tracking framework using PyTorch 1.7.0 and Python 3.8.10. All experiments are conducted on a server with two RTX A5000 GPUs. The training data include LaSOT [3], GOT-10K [39], COCO2017 [40], and TrackingNet [41]. Consistent with previous works, we also remove 1k sequences from the GOT-10K training set that are prohibited by VOT2019 challenge. The size of search images and templates are 256×256 pixels and 128×128 pixels respectively. Each epoch is composed of 60,000 sample pairs, and the total number of epochs is set to be 500. The model is trained utilizing the AdamW optimizer [42], with a weight decay set to 10^{-4} . The initial learning rate for the backbone is 10^{-5} , while the initial learning rate for the remaining parts, excluding the backbone, is set to 10^{-4} . The learning rate undergoes a decay by a factor of 10 at the 400th epoch.

Testing phase. In line with most Siamese trackers, target features are only extracted during the initialization phase and are not updated in subsequent frames. For each following video frame, we crop the search region based on the previous frame's target position and obtain a predicted location through a single forward pass. To incorporate location priors into the tracking process, we utilize a Hanning window penalty [43] following the common practice [26,44]. In a specific approach, we conduct multiplication of the classification map P with a Hanning window of identical dimensions. Subsequently, the tracking result is determined by selecting the highest-scoring box after this multiplication process.

Table 1

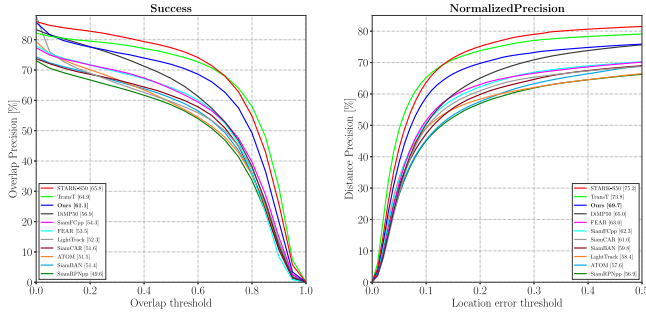
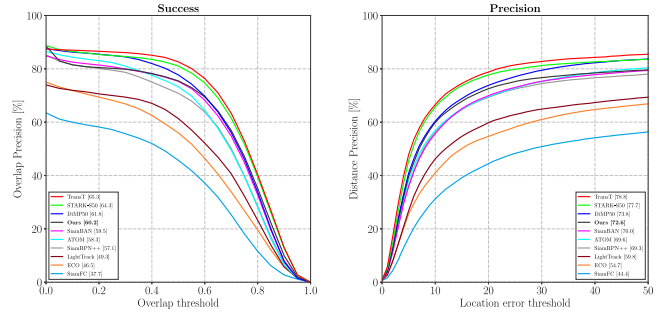
Results on GOT-10K. Top-3 results of each dimension (row) are colored in red, blue and green, respectively.

		Non-realtime					Realtime				
Trackers		SiamRPN++ [16]	SiamFC++ [20]	DiMP-50 [23]	TransT [26]	STARTK-S50 [2]	SiamFC [14]	ATOM [22]	LightTrack [4]	FEAR [10]	Ours
GOT10K	SR ₇₅ ↑	32.5	47.9	49.2	60.9	61.2	9.8	40.2	—	—	57.3
	AO ↑	51.8	59.5	61.1	67.1	67.9	34.8	55.6	61.1	61.9	63.6

Table 2

Results on TrackingNet test set. Top-3 results of each dimension (row) are colored in red, blue and green, respectively [2,4,14,16,20,22,23,26,47].

		Non-realtime					Realtime				
Trackers		SiamRPN++ [16]	SiamFC++ [20]	DiMP-50 [23]	TransT [26]	STARTK-S50 [2]	SiamFC [14]	ATOM [22]	LightTrack [4]	E.T.Track [47]	Ours
TrackingNet	P _{norm} (%) ↑	80.0	80.0	80.1	86.7	85.1	66.3	77.1	77.9	80.3	83.8
	AUC (%) ↑	73.3	75.4	74.0	81.4	80.3	57.1	70.3	72.5	74.9	78.1

**Fig. 5.** Comparitions on LaSOT test set.**Fig. 6.** Success and precision plots on NFS benchmark.

4.2. Evaluation

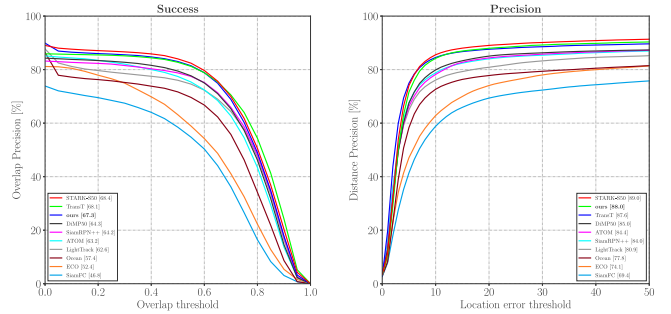
In this section, we conduct a comparative analysis between our proposed MobileTrack and current state-of-the-art methods across 6 benchmark datasets: NFS [45], LaSOT [3], TrackingNet [41], GOT-10K [39], UAV123 [46], and VOT2020 [5].

On LaSOT. LaSOT stands as a challenging benchmark for large-scale long-term tracking, encompassing a set of 280 testing videos. The dataset poses significant challenges, particularly in the context of very long sequences, with an average sequence length of 2500 frames. Therefore, robustness plays an essential role in achieving high performance on LaSOT. We compare the result of the MobileTrack with previous SOTA trackers in Fig. 5. Our proposed model demonstrates high robustness, achieving an AUC score of 61.1%, indicating a significant improvement of 4.2% compared to the well-established DiMP tracker. Additionally, when contrasted with the lightweight mobile architecture of LightTrack, our model showcases a remarkable 8.8% enhancement in success score, accompanied by improved speed.

On GOT-10K. GOT-10K benchmark presents a comprehensive evaluation platform that encompasses an array of objects, their diverse deformations, and occlusions. Our model undergoes training based on this standardized protocol, followed by submission to the official evaluation server for meticulous scrutiny and assessment of its performance. In Table 1, we present a comparative evaluation of our tracker against 9 other state-of-the-art trackers. Our MobileTrack tracker outperforms both LightTrack [4] and FEAR [10] with higher accuracy while maintaining an impressive speed.

On TrackingNet. The TrackingNet benchmark, which contains 511 sequences covering diverse target classes, presents a comprehensive testing platform. We submit the outputs of our tracker to the official online evaluation server and present the AUC and P_{norm} results. The outcomes are detailed in Table 2, showcasing the exceptional performance of our MobileTrack tracker, surpassing all other real-time tracking methods.

On NFS. The assessment of the provided tracker is carried out on the version of the NFS dataset captured at a frame rate of 30 fps.

**Fig. 7.** Success and precision plots on UAV123 benchmark.

This dataset includes videos with challenging scenarios involving high-speed motion of objects. We compare the result of the MobileTrack with previous SOTA trackers in Fig. 6. Our method achieves an AUC score of 60.2%. This score represents a significant improvement over lightweight algorithms like LightTrack [4].

On UAV123. The UAV123 stands as an aerial video benchmark and dataset, comprising 123 sequences captured from a low-altitude aerial perspective. These benchmarks are designed for evaluating the performance of UAVs in real-time scenarios. All sequences within the dataset are meticulously annotated with upright bounding boxes. Notably, the objects featured in the dataset commonly encounter challenges such as fast motion, large-scale variation, and occlusions, rendering tracking tasks particularly demanding. As illustrated in Fig. 7, our tracker achieves an AUC score of 67.3 and a Prec score of 88.0. These results consistently surpass those obtained by recent competitive Siamese trackers, including SiamRPN++ [16], LightTrack [4], and DiMP [23].

On VOT2020. In the VOT-ST2020 evaluation, trackers are subjected to a comprehensive analysis that includes measurements of accuracy, robustness, and the expected average overlap (EAO). For accuracy assessment, we employ a weighted calculation of the mean overlap between the ground truth and predicted targets within subsequences specified by anchors. This approach allows for a nuanced evaluation of

Table 3

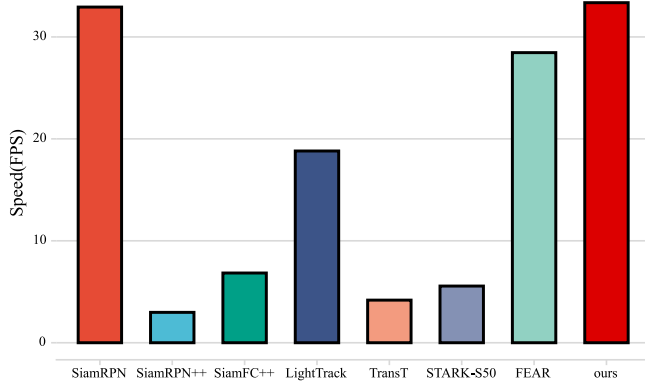
Results on VOT-ST2020. Top-3 results of each dimension (row) are colored in red, blue and green, respectively.

		Non-realtime		Realtime				
Trackers		DiMP-50 [23]	STARTK-S50 [2]	KCF [12]	SiamFC [14]	ATOM [22]	LightTrack [4]	Ours
VOT-ST2020	EAO $\uparrow$	0.274	0.280	0.154	0.179	0.271	0.242	0.265
	Accuracy $\uparrow$	0.457	0.478	0.407	0.418	0.462	0.422	0.476
	Robustness $\uparrow$	0.740	0.799	0.432	0.502	0.734	0.689	0.684

Table 4

Ablation study on LaSOT [3] to investigate the impact of each component in terms of AUC (%) and FPS.

#	Component	AUC $\uparrow$	#Parameters (M) $\downarrow$	FLOPs (G) $\downarrow$	FPS $\uparrow$
1	baseline	57.9	4.49	2.11	29.86
2	+coordinated perception-aware fusion module	61.3	4.50	2.12	29.15
3	+shared weight classification and regression head	61.1	2.59	1.63	33.36

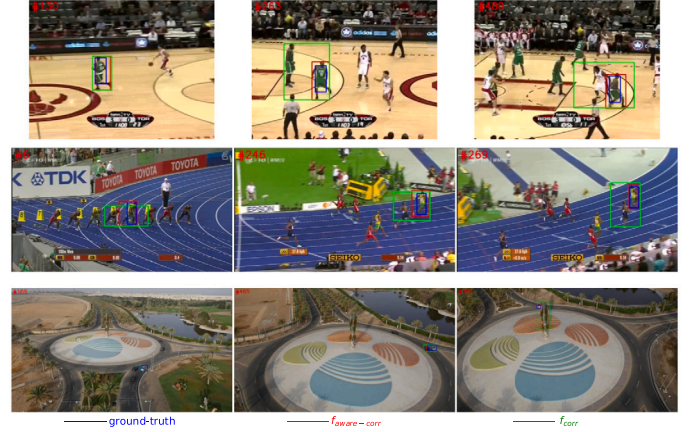
**Fig. 8.** Run-time speed on NVIDIA Jetson TX2.

tracking precision. The evaluation also considers robustness, indicating the proportion of frames before trackers encounter failure on average. Additionally, the expected average overlap (EAO) serves as a holistic metric, combining accuracy and robustness, providing a comprehensive measure of overall tracking performance. The results are shown in Table 3. The results demonstrate that our model achieves a remarkable accuracy score of 0.476, which is almost equivalent to that of STARK-S50 [2]. Moreover, our model's EAO of 0.265 outperforms LightTrack by a significant margin of 9.5%. These findings highlight the effectiveness of our approach in improving the performance of object tracking algorithms.

Speed. We deploy multiple algorithms to the NVIDIA Jetson TX2 edge device to verify their inference speed. Fig. 8, summarizes the running speed of these algorithms on edge platforms. Most algorithms fail to meet real-time requirements, with only SiamRPN [15] and our approach demonstrating efficient performance. The real-time inference speed enables the deployment of our approach on resource-constrained platforms, such as drones, which frequently utilize edge chipsets.

4.3. Ablation study

To ascertain the effectiveness of our proposed approach, we analyze the influence of its distinct components on the LaSOT benchmark, as outlined in Table 4. The baseline model(#1) employs MobileViT-S as the backbone with the final stage removed and features from the last layer are outputted with a stride of 16. The neck block utilized in this study is the pixel-wise fusion block presented in FEAR. Additionally, the corner head network is employed to estimate the probability distribution of the box corners. Results obtained from the baseline exhibit an AUC of 57.9 and FPS of 29.86. In experiment #2, the coordinated perception-aware fusion module is substituted for the pixel-wise fusion block. This replacement resulted in a 3% improvement in AUC with

**Fig. 9.** A comparative analysis is conducted, evaluating the respective final tracking outcomes obtained through f_{corr} and $f_{aware-corr}$. It is observed that $f_{aware-corr}$ exhibits superior tracking performance, showcasing enhanced capabilities in tracking accuracy and reliability compared to f_{corr} .

negligible degradation of speed. This indicates that the utilization of the coordinated perception-aware fusion module results in learned features that possess stronger discriminative ability while effectively encoding the position and appearance of objects in the tracker.

As shown in Fig. 9, we visualize the ground-truth bounding boxes and the tracking outcomes obtained separately through f_{corr} and $f_{aware-corr}$. Through this comparison, it becomes evident that the feature map $f_{aware-corr}$, incorporating both channel and positional information, exhibits a heightened focus on the target location. Particularly noteworthy is its enhanced concentration on the target location even in the presence of interfering objects. This characteristic proves beneficial in scenarios involving occlusions or distractors, where $f_{aware-corr}$ demonstrates a superior capability to maintain focus on the intended target position. Therefore, utilizing channel attention and position attention operations can result in more discriminative features that improve tracking performance.

Moreover, the instantiation of the suggested shared weight classification and regression head (#3) leads to a 0.2% reduction in AUC but a noteworthy enhancement in FPS, thus showcasing the efficacy of this module. The integration of the coordinated perception-aware fusion module and the shared weight classification regression tracking head yields a considerable augmentation in both tracking accuracy and inference speed.

5. Conclusion

In this paper, we present a novel and efficient vision object tracking method. Our approach achieves an optimal trade-off between accuracy and speed, rendering it a highly promising solution for practical

applications that demand both efficient and precise visual tracking capabilities. However, the feature fusion approach proposed by MobileTrack still falls short, unable to address tracking tasks in complex scenarios. Exploring more effective methods for relationship modeling to obtain distinctive features remains a significant research challenge. Despite this, our MobileTrack framework is capable of real-time inference on diverse resource-restricted platforms, contributing to the provision of more efficient and accurate solutions for various edge devices and application scenarios.

CRedit authorship contribution statement

Jiang Zhai: Writing – original draft. **Zinan Cheng:** Resources. **Wenkang Zhang:** Validation. **Dejun Zhu:** Writing – review & editing. **Wankou Yang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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